

Generel opstilling af det duale LP

Maksimering af det primale LP

Det primale LP

max.	$c_1x_1 + c_2x_2 + c_3x_3 + c_4x_4$	Dual variabel
s.t.	$a_1x_1 + a_2x_2 + a_3x_3 + a_4x_4 \leq a_0$	y_a
	$b_1x_1 + b_2x_2 + b_3x_3 + b_4x_4 \geq b_0$	y_b
	$d_1x_1 + d_2x_2 + d_3x_3 + d_4x_4 = d_0$	y_d
	$x_1 \geq 0, x_2 \geq 0, x_3 \leq 0, x_4 \text{ fri}$	

Det duale LP

min.	$a_0y_a + b_0y_b + d_0y_d$	Primal variabel
s.t.	$a_1y_a + b_1y_b + d_1y_d \geq c_1$	x_1
	$a_2y_a + b_2y_b + d_2y_d \geq c_2$	x_2
	$a_3y_a + b_3y_b + d_3y_d \leq c_3$	x_3
	$a_4y_a + b_4y_b + d_4y_d = c_4$	x_4
	$y_a \geq 0, y_b \leq 0, y_d \text{ fri}$	

Generelle relationer

Primal	Dual
Obj.: Max.	Obj.: Min.
Begr.	Variabel
$\leq a_0$	$y_a \geq 0$
$\geq b_0$	$y_b \leq 0$
$= d_0$	$y_d \text{ fri}$
Variabel	Begr.
$x_1 \geq 0$	$\geq c_1$
$x_2 \geq 0$	$\geq c_2$
$x_3 \leq 0$	$\leq c_3$
$x_4 \text{ fri}$	$= c_4$

Minimering af det primale LP

Det primale LP

min.	$c_1x_1 + c_2x_2 + c_3x_3 + c_4x_4$	Dual variabel
s.t.	$a_1x_1 + a_2x_2 + a_3x_3 + a_4x_4 \leq a_0$	y_a
	$b_1x_1 + b_2x_2 + b_3x_3 + b_4x_4 \geq b_0$	y_b
	$d_1x_1 + d_2x_2 + d_3x_3 + d_4x_4 = d_0$	y_d
	$x_1 \geq 0, x_2 \geq 0, x_3 \leq 0, x_4 \text{ fri}$	

Det duale LP

max.	$a_0y_a + b_0y_b + d_0y_d$	Primal variabel
s.t.	$a_1y_a + b_1y_b + d_1y_d \leq c_1$	x_1
	$a_2y_a + b_2y_b + d_2y_d \leq c_2$	x_2
	$a_3y_a + b_3y_b + d_3y_d \geq c_3$	x_3
	$a_4y_a + b_4y_b + d_4y_d = c_4$	x_4
	$y_a \leq 0, y_b \geq 0, y_d \text{ fri}$	

Generelle relationer

Primal	Dual
Obj.: Min.	Obj.: Max.
Begr.	Variabel
$\leq a_0$	$y_a \leq 0$
$\geq b_0$	$y_b \geq 0$
$= d_0$	$y_d \text{ fri}$
Variabel	Begr.
$x_1 \geq 0$	$\leq c_1$
$x_2 \geq 0$	$\leq c_2$
$x_3 \leq 0$	$\geq c_3$
$x_4 \text{ fri}$	$= c_4$